

MATH 449 Project

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State the problem and describe the data set

Introduction

Wine quality is influenced by several physicochemical properties, such as acidity, sugar, sulfur dioxide, sulphates, density, and alcohol content. In this project, I use the WineQuality-Red data set from the UC Irvine Machine Learning Repository to study which chemical measurements are related to higher wine quality.

The original response variable, `quality`, is measured on a scale from 0 to 10. To use logistic regression, I converted this response into a binary variable. Wines with a quality score greater than 6 are classified as high quality, while wines with a quality score of 6 or below are classified as not high quality.

The goal of this project is to build a logistic regression model that predicts the probability that a red wine is high quality based on its physicochemical measurements. I also compare different link functions, evaluate classification performance, and determine which variables appear to be important predictors of high-quality wine.

Fit a logistic regression model with all predictors

I fit a logistic regression model using all available physicochemical variables as predictors. The response variable is whether a wine is classified as high quality (`quality > 6`). The predictors include measurements related to acidity, sulfur compounds, density, sugar content, pH, sulphates, and alcohol content.

```
df <- read.csv("~/MATH449_R/449_Scripts/wine+quality/winequality-red.csv",
               header = TRUE, sep = ";")
variable.names(df)
```

```
## [1] "fixed.acidity"      "volatile.acidity"   "citric.acid"
## [4] "residual.sugar"     "chlorides"          "free.sulfur.dioxide"
## [7] "total.sulfur.dioxide" "density"            "pH"
## [10] "sulphates"         "alcohol"            "quality"
```

```
#Fit the model
model.logit <- glm((quality > 6) ~ fixed.acidity + volatile.acidity + citric.acid +
                  residual.sugar + chlorides + free.sulfur.dioxide +
                  total.sulfur.dioxide + density + pH + sulphates + alcohol,
                  family = binomial, data = df)

summary(model.logit)
```

```
##
## Call:
## glm(formula = (quality > 6) ~ fixed.acidity + volatile.acidity +
##      citric.acid + residual.sugar + chlorides + free.sulfur.dioxide +
##      total.sulfur.dioxide + density + pH + sulphates + alcohol,
##      family = binomial, data = df)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      2.428e+02  1.081e+02   2.247 0.024660 *
## fixed.acidity      2.750e-01  1.253e-01   2.195 0.028183 *
## volatile.acidity  -2.581e+00  7.843e-01  -3.291 0.000999 ***
## citric.acid        5.678e-01  8.385e-01   0.677 0.498313
## residual.sugar     2.395e-01  7.373e-02   3.248 0.001163 **
## chlorides        -8.816e+00  3.365e+00  -2.620 0.008788 **
## free.sulfur.dioxide 1.082e-02  1.223e-02   0.884 0.376469
## total.sulfur.dioxide -1.653e-02  4.894e-03  -3.378 0.000731 ***
## density          -2.578e+02  1.104e+02  -2.335 0.019536 *
## pH                2.242e-01  9.984e-01   0.225 0.822327
## sulphates         3.750e+00  5.416e-01   6.924 4.39e-12 ***
## alcohol           7.533e-01  1.316e-01   5.724 1.04e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1269.92  on 1598  degrees of freedom
## Residual deviance:  870.86  on 1587  degrees of freedom
## AIC: 894.86
##
## Number of Fisher Scoring iterations: 6
```

The full logistic regression model identified several statistically significant predictors of wine quality. Variables such as volatile acidity, residual sugar, chlorides, total sulfur dioxide, sulphates, and alcohol showed strong associations with the probability that a wine is classified as high quality. Some variables, including citric acid, free sulfur dioxide, and pH, were not statistically significant in the full model.

Select the best subset of variables. Perform a diagnostic on the best model. Perform all possible inferences you can think about. Perform goodness of fit tests. Find confidence intervals for multiplicative effects.

To simplify the model and improve the interpretability, I removed predictors that weren't statistically significant in the full logistic regression model. To simplify the model, I removed predictors that were not statistically significant in the full logistic regression model.

#Removed insignificant variables based on the fitted full model

```
model1.logit <- glm((quality > 6) ~ fixed.acidity + volatile.acidity +
                    residual.sugar + chlorides + total.sulfur.dioxide + density +
                    sulphates + alcohol,
                    family = binomial, data = df)

summary(model1.logit)
```

```
##
## Call:
## glm(formula = (quality > 6) ~ fixed.acidity + volatile.acidity +
##      residual.sugar + chlorides + total.sulfur.dioxide + density +
##      sulphates + alcohol, family = binomial, data = df)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      2.268e+02  9.163e+01   2.475 0.013336 *
## fixed.acidity      2.812e-01  8.029e-02   3.502 0.000462 ***
## volatile.acidity   -2.913e+00  6.467e-01  -4.504 6.66e-06 ***
## residual.sugar      2.328e-01  7.009e-02   3.322 0.000893 ***
## chlorides         -8.441e+00  3.259e+00  -2.590 0.009593 **
## total.sulfur.dioxide -1.360e-02  3.447e-03  -3.946 7.95e-05 ***
## density           -2.409e+02  9.202e+01  -2.618 0.008835 **
## sulphates          3.699e+00  5.287e-01   6.997 2.62e-12 ***
## alcohol            7.823e-01  1.120e-01   6.983 2.88e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1269.92  on 1598  degrees of freedom
## Residual deviance:  872.08  on 1590  degrees of freedom
## AIC: 890.08
##
## Number of Fisher Scoring iterations: 6
```

```
anova(model.logit, model1.logit, test='Chisq')
```

```
## Analysis of Deviance Table
##
## Model 1: (quality > 6) ~ fixed.acidity + volatile.acidity + citric.acid +
##      residual.sugar + chlorides + free.sulfur.dioxide + total.sulfur.dioxide +
##      density + pH + sulphates + alcohol
## Model 2: (quality > 6) ~ fixed.acidity + volatile.acidity + residual.sugar +
##      chlorides + total.sulfur.dioxide + density + sulphates +
##      alcohol
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      1587      870.86
## 2      1590      872.08 -3   -1.2116   0.7502
```

I then compared the full and reduced models using a likelihood ratio test.

H_0 : The reduced and full models fit the data equally well

H_a : The models fit the data differently

The likelihood ratio test gave us a p-value of 0.7502. Since this value is greater than $\alpha = 0.05$, we fail to reject the null hypothesis. This means there's not enough evidence that the removed variables (citric.acid, free.sulfur.dioxide, and pH) significantly improve the model fit.

```
AIC(model.logit)
```

```
## [1] 894.8644
```

```
AIC(model1.logit)
```

```
## [1] 890.076
```

The AIC was also used to compare the two models. The reduced model had a smaller AIC value than the full model (890.076 vs. 894.8644). Since smaller AIC values indicate a better balance between model fit and model complexity, the reduced model is what we will use for the rest of the analysis.

After selecting the reduced model, I evaluated overall model fit and looked at the multiplicative effects of the predictors using odds ratios and confidence intervals.

```
# Goodness of fit test  
model1.logit$deviance
```

```
## [1] 872.076
```

```
model1.logit$df.residual
```

```
## [1] 1590
```

```
1 - pchisq(model1.logit$deviance, model1.logit$df.residual)
```

```
## [1] 1
```

```
# Multiplicative effects (AKA odds ratio)  
exp(coef(model1.logit))
```

##	(Intercept)	fixed.acidity	volatile.acidity
##	2.995201e+98	1.324678e+00	5.432158e-02
##	residual.sugar	chlorides	total.sulfur.dioxide
##	1.262184e+00	2.158766e-04	9.864922e-01
##	density	sulphates	alcohol
##	2.281752e-105	4.039619e+01	2.186392e+00

```
# Confidence intervals for multiplicative effects  
exp(confint(model1.logit))
```

```
## Waiting for profiling to be done...
```

##		2.5 %	97.5 %
##	(Intercept)	5.580631e+20	9.590413e+176
##	fixed.acidity	1.132372e+00	1.551774e+00
##	volatile.acidity	1.491835e-02	1.885536e-01
##	residual.sugar	1.093693e+00	1.443432e+00
##	chlorides	1.680634e-07	5.642012e-02
##	total.sulfur.dioxide	9.795519e-01	9.928975e-01
##	density	3.194303e-184	2.548508e-27
##	sulphates	1.432648e+01	1.151379e+02
##	alcohol	1.758834e+00	2.730749e+00

H_0 : The logistic regression model fits the data adequately

H_a : The logistic regression model does not fit the data adequately

The goodness-of-fit test produced a p-value close to 1, so we fail to reject the null hypothesis. Therefore, there is no evidence that the logistic regression model does not fit the data adequately. Overall, the reduced model appears to provide a good fit for the data.

The multiplicative effects show that several physicochemical variables are strongly associated with wine quality.

For alcohol, we are 95% confident that a one-unit increase in alcohol multiplies the odds of a wine being classified as high quality by between 1.758834 and 2.730749, holding all other variables constant. This suggests that wines with higher alcohol content are more likely to be classified as high quality.

For sulphates, we are 95% confident that a one-unit increase in sulphates multiplies the odds of high-quality wine by between 14.32648 and 115.1379, holding all other variables constant. This shows a strong positive association with wine quality.

In contrast, volatile acidity and chlorides had negative multiplicative effects, indicating that larger values of these variables decrease the odds that a wine is classified as high quality.

Overall, the confidence intervals suggest that several physicochemical measurements have meaningful effects on wine quality classification.

Use the new model to make predictions.

```
pred.prob <- predict(model1.logit, type = "response")
head(pred.prob)
```

```
##           1           2           3           4           5           6
## 0.007680065 0.008602474 0.012139372 0.070695593 0.007680065 0.007834548
```

```
summary(pred.prob)
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 0.0005455 0.0155443 0.0455076 0.1357098 0.1799477 0.9894081
```

The predicted probabilities ranged from 0.0005455 to 0.9894081, meaning there is large variation in the probability that wines are classified as high quality. Some wines had probabilities close to 0, while others had probabilities close to 1. The mean predicted probability was 0.1357098.

Use different `pi_0` as a cut-off point and create a confusion table.

Different cutoff values were used to classify wines as high quality or not high quality based on the predicted probabilities from the logistic regression model.

```

y <- as.numeric(df$quality > 6)

# $\pi_0 = 0.5$ 
predicted1 <- as.numeric(fitted(model1.logit) > 0.5)

xtabs(~ y + predicted1)

```

```

##      predicted1
## y          0    1
## 0 1336    46
## 1  145    72

```

```

#  $\pi_0 = \text{sample proportion of high-quality wines}$ 
prop <- sum(y)/nrow(df)

predicted2 <- as.numeric(fitted(model1.logit) > prop)

xtabs(~ y + predicted2)

```

```

##      predicted2
## y          0    1
## 0 1078   304
## 1   39   178

```

```

#  $\pi_0 = 0.3$ 
predicted3 <- as.numeric(fitted(model1.logit) > 0.3)

xtabs(~ y + predicted3)

```

```

##      predicted3
## y          0    1
## 0 1250   132
## 1   85   132

```

Using $\pi_0 = 0.5$, the model was more strict when classifying wines as high quality. Most wines were classified as not high quality, and the model correctly identified 72 high-quality wines.

Using $\pi_0 = \hat{\pi}$, the sample proportion of high-quality wines, the model classified more wines as high quality. This increased the number of correctly identified high-quality wines to 178, but also increased the number of false positives.

Using $\pi_0 = 0.3$, the model became less strict than when using $\pi_0 = 0.5$, which gave us more wines being classified as high quality.

Perform visualization of data and models.

To better understand the relationship between alcohol content and wine quality, I visualized the fitted logistic regression model along with estimated confidence bounds.

```

y <- as.numeric(df$quality > 6)

plot(jitter(y, 0.1) ~ alcohol,
     xlab = "Alcohol",
     ylab = "P(High Quality Wine)",
     data = df)

data.plot <- data.frame(
  alcohol = seq(min(df$alcohol), max(df$alcohol), length = 100),
  fixed.acidity = mean(df$fixed.acidity),
  volatile.acidity = mean(df$volatile.acidity),
  residual.sugar = mean(df$residual.sugar),
  chlorides = mean(df$chlorides),
  total.sulfur.dioxide = mean(df$total.sulfur.dioxide),
  density = mean(df$density),
  sulphates = mean(df$sulphates)
)

lp <- predict(model1.logit, newdata = data.plot, se.fit = TRUE)

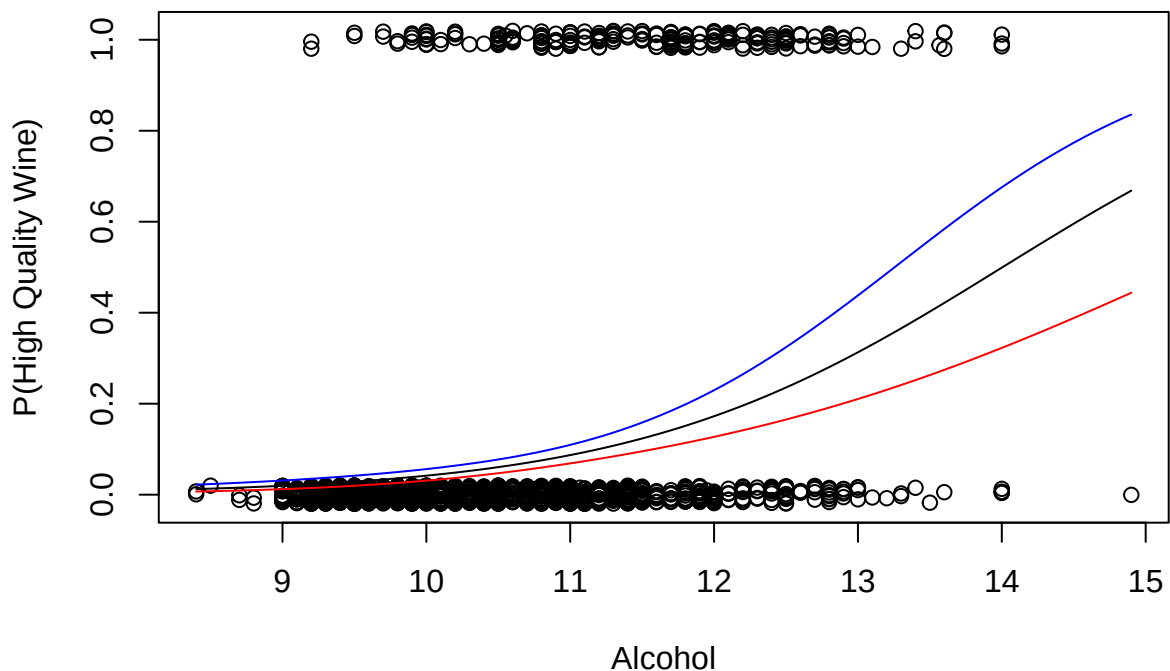
pred.prob <- exp(lp$fit)/(1 + exp(lp$fit))

LB <- lp$fit - qnorm(0.975)*lp$se.fit
UB <- lp$fit + qnorm(0.975)*lp$se.fit

LB.p <- exp(LB)/(1 + exp(LB))
UB.p <- exp(UB)/(1 + exp(UB))

lines(data.plot$alcohol, pred.prob)
lines(data.plot$alcohol, LB.p, col = "red")
lines(data.plot$alcohol, UB.p, col = "blue")

```



From the plot we can see that as alcohol content increases, the probability of a wine being classified as high quality also increases. The black curve shows the estimated probability from the logistic regression model.

The red and blue curves are confidence bounds for the estimated probabilities. The curves spread out more for larger alcohol values, showing that the model becomes a little less certain about predictions at higher alcohol levels.

Plot the ROC curve, find AUC, and the best cutoff point for classification.

ROC analysis was used to evaluate the classification performance of the logistic regression model across different cutoff values.

```
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.
```

```
##
```

```
## Attaching package: 'pROC'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

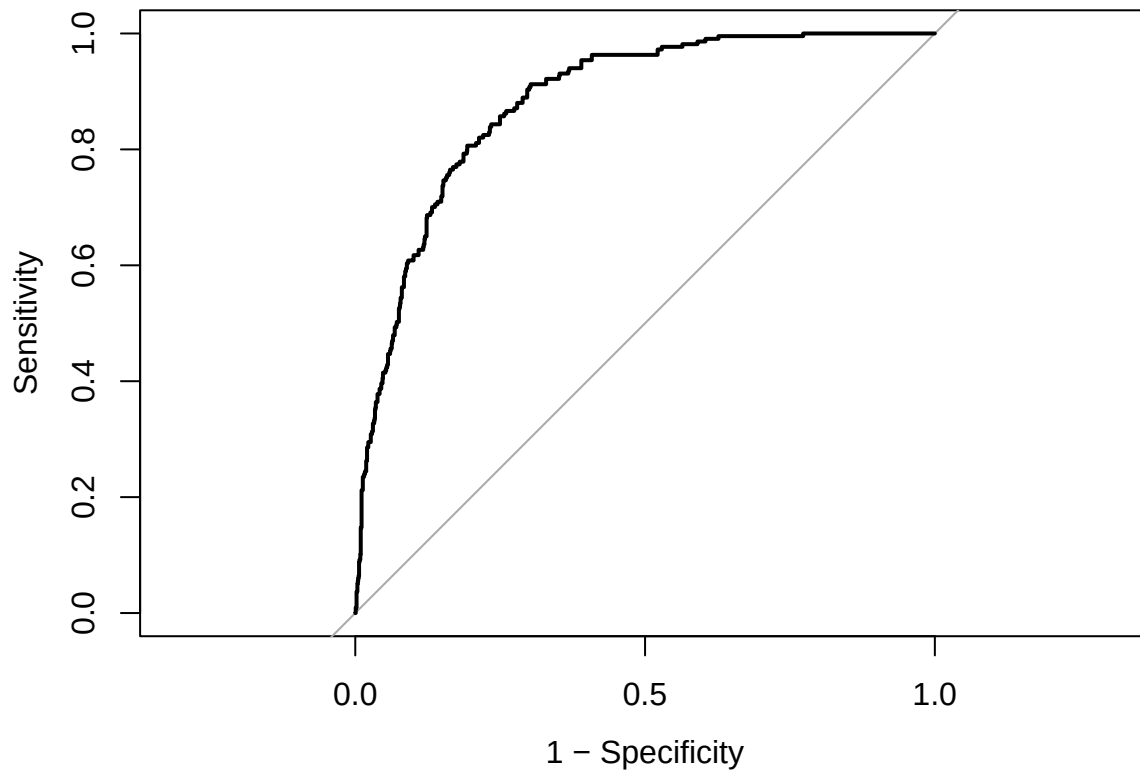
```
##      cov, smooth, var
```

```
rocplot <- roc(y ~ fitted(model1.logit), data = df)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
plot.roc(rocplot, legacy.axes = TRUE)
```

```
auc(rocplot)
```

```
## Area under the curve: 0.882
```

The ROC curve shows us the tradeoff between sensitivity and specificity for different classification cutoff values. The area under the curve was $AUC = 0.882$, which shows us that the logistic regression model has a strong ability to distinguish between high-quality and low-quality wines.

Since the AUC value is relatively close to 1, the model has good overall classification performance.

Perform LOOCV and k-fold cross-validation.

I used Cross-validation methods to evaluate how well the logistic regression model performs on new data.

```
#LOOCV

pihat <- vector(length = nrow(df))
for(i in 1:nrow(df))
{
  pihat[i] <-
    predict(update(model1.logit, subset = -i),
            newdata = df[i,],
            type = "response")
}

yy <- as.numeric(df$quality > 6)
prop <- sum(yy)/nrow(df)
```

```
yhat <- as.numeric(pihat > prop)
xtabs(~ yy + yhat)
```

```
##      yhat
## yy      0      1
##    0 1078   304
##    1    41   176
```

```
# k-fold cross validation
y <- as.numeric(df$quality > 6)
library(lattice)
library(DAAG)
cv.binary(model1.logit) # here pi_0 = 0.5
```

```
##
## Fold:  5 9 1 8 3 6 2 10 4 7
## Internal estimate of accuracy = 0
## Cross-validation estimate of accuracy = 0
```

```
library(boot)
```

```
##
## Attaching package: 'boot'
```

```
## The following object is masked from 'package:lattice':
##
##      melanoma
```

```
prop <- sum(y)/nrow(df)
cost <- function(r, pi = 0) mean(abs(r - pi) > prop)

set.seed(123)
out <- cv.glm(df, model1.logit, cost, K = 10)
names(out)
```

```
## [1] "call" "K" "delta" "seed"
```

```
out$delta
```

```
## [1] 0.3220763 0.3232582
```

Using LOOCV, the model correctly classified many wines as either high quality or not high quality. The LOOCV results were similar to the earlier confusion tables. This suggests that the model gives us consistent predictions.

For the 10-fold cross-validation, the estimated prediction error rates were 0.3220763 and 0.3232582. This tells us that the model misclassified observations about 32% of the time during cross-validation.

Try the probit link and the identity links to model data.

To compare alternative link functions, I fit both a probit model and an identity-link binomial model using the same predictor variables as the reduced logistic regression model.

```
# Probit link
model.probit <- glm((quality > 6) ~ fixed.acidity + volatile.acidity +
                    residual.sugar + chlorides + total.sulfur.dioxide +
                    density + sulphates + alcohol,
                    family = binomial(link = "probit"), data = df)
summary(model.probit)
```

```
##
## Call:
## glm(formula = (quality > 6) ~ fixed.acidity + volatile.acidity +
##     residual.sugar + chlorides + total.sulfur.dioxide + density +
##     sulphates + alcohol, family = binomial(link = "probit"),
##     data = df)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.205e+02  5.068e+01   2.378  0.01741 *
## fixed.acidity    1.442e-01  4.388e-02   3.286  0.00102 **
## volatile.acidity -1.567e+00  3.459e-01  -4.529  5.91e-06 ***
## residual.sugar    1.237e-01  3.905e-02   3.168  0.00154 **
## chlorides       -4.403e+00  1.644e+00  -2.678  0.00740 **
## total.sulfur.dioxide -7.457e-03  1.861e-03  -4.006  6.17e-05 ***
## density         -1.283e+02  5.087e+01  -2.522  0.01168 *
## sulphates        2.039e+00  2.974e-01   6.858  6.96e-12 ***
## alcohol         4.297e-01  6.178e-02   6.955  3.52e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1269.92  on 1598  degrees of freedom
## Residual deviance:  868.86  on 1590  degrees of freedom
## AIC: 886.86
##
## Number of Fisher Scoring iterations: 7
```

The probit model gave us similar results to the logistic regression model. The model had an AIC value of 886.86, which was slightly smaller than the logistic regression model AIC of 890.08. This tells me that the probit link may give us a slightly better fit for this data set.

```
# Identity link
model.identity <- try(
  glm((quality > 6) ~ fixed.acidity + volatile.acidity + residual.sugar +
      chlorides + total.sulfur.dioxide + density + sulphates + alcohol,
      family = binomial(link = "identity"), data = df)
)
```

```
## Error : no valid set of coefficients has been found: please supply starting values
```

```
model.identity
```

```
## [1] "Error : no valid set of coefficients has been found: please supply starting values\n"
## attr(,"class")
## [1] "try-error"
## attr(,"condition")
## <simpleError: no valid set of coefficients has been found: please supply starting values>
```

The identity-link model didn't converge and printed the error message "no valid set of coefficients has been found." This tells me that the identity link function isn't appropriate for modeling this data.

Which model works better for this data?

Overall, the probit model performed slightly better for this data because it gave us a smaller AIC value and slightly smaller residual deviance than the logistic regression model. But both models produced similar results and similar classification performance. The identity-link model didn't fit successfully and therefore isn't appropriate for this analysis.

11.If you have grouped data, use the methods for contingency tables to analyze the data (Chi sq test, G^2 , and so on if applicable).

The WineQuality-Red data set isn't grouped in contingency table data, so the contingency table methods like Chi-square test and likelihood ratio (G^2) test aren't applicable for this analysis.

Conclusion

In this project, logistic regression was used to model the probability that a red wine is classified as high quality based on physicochemical measurements. After comparing the full model and reduced model, I found that the reduced logistic regression model was a better choice because it gave me a similar model performance while using fewer predictors.

Several variables showed meaningful relationships with wine quality. Specifically alcohol and sulphates had strong positive associations with high-quality wine, while volatile acidity and chlorides showed negative associations. The ROC analysis gave me an AUC value of 0.882, showing good classification performance.

The cross-validation results suggested that the model produced pretty stable predictions on new data. Also, the probit link function produced results similar to the logistic regression model and slightly improved the AIC value, while the identity-link model failed to converge and wasn't appropriate for this data set.

Overall, the analysis demonstrates that physicochemical measurements can be useful predictors of wine quality classification.

References

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Wine Quality [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C56S3T>.